# Tutorial 3
# Sets and Functions

# Sets

# Power sets

How many elements does each of these sets have where $a$ and $b$ are distinct elements?

**a)** $\mathcal{P}(\{a, b, \{a, b\}\})$

**b)** $\mathcal{P}(\{\emptyset, a, \{a\}, \{\{a\}\}\})$

**c)** $\mathcal{P}(\mathcal{P}(\emptyset))$

Let A be a set, and the elements of A be sets.

Define $\bigcup A = \{x \mid \exists y \in A, x \in y\}$.

1) Calculate $\bigcup \{\{a,b,c\},\{a,d,e\},\{a,f\}\}$;

2) Calculate $\bigcup P(A)$

3) Whether $P(\bigcup A) = A$?

$A = \{\{a,b\}, \{b,c\}\}$

$\bigcup A = \{a,b,c\}$

$A = \{\{\{a\},\{b\}\}, \{b,c\}\}$

$\bigcup A = \{\{a\},\{b\}\} \cup \{b,c\}$

$= \{\{a\},\{b\}, c\}$

$$\bigcup A = \{x \mid \exists y \in A, x \in y\}.$$

Answer:

1). Calculate $\bigcup\{\{a,b,c\},\{a,d,e\},\{a,f\}\}$;

Solution: $\bigcup\{\{a,b,c\},\{a,d,e\},\{a,f\}\}$

$$= \{a,b,c\} \bigcup \{a,d,e\} \bigcup \{a,f\}$$

$$= \{a,b,c,d,e,f\}$$

(2) $A = \{\{a,b,c\}, \{a,d,e\}, \{a,f\}\}$

$P(A) = \{\phi, \{\{a,b,c\}\}, \{\{a,d,e\}\}, \{\{a,f\}\}, \{\{a,b,c\},\{a,d,e\}\}, \{\{a,b,c\},\{a,f\}\}, \{\{a,d,e\},\{a,f\}\}, A\}$

$UP(A) = \{\{a,b,c\}, \{a,d,e\}, \{a,f\}\}$

$$\bigcup A = \{x \mid \exists y \in A, x \in y\}.$$

3). Whether $P(\bigcup A) = A$?

Answer: No!

Counter example:

$$A = \{\{a,b,c\},\{a,d,e\},\{a,f\}\}$$

$$\bigcup A = \{a,b,c,d,e,f\}$$

$$\{a,e\} \in P(\bigcup A), \quad \text{but} \quad \{a,e\} \notin A$$

$$\text{so } P(\bigcup A) \neq A$$

- Let A and B be sets, prove

$$A \cap B = \emptyset \leftrightarrow A \subseteq \bar{B}$$

*We split task into two subtasks.*

*The* ***first*** *is to prove that* $A \cap B = \emptyset \rightarrow A \subseteq \overline{B}$.

*By way of contradiction, suppose* $A \cap B = \emptyset$ *and* $A \nsubseteq \overline{B}$.

*If* $A \nsubseteq \overline{B}$, *we can find an* $x$, *such that* $x \in A$ *and* $x \notin \overline{B}$.

*But,*

$x \in A \wedge x \notin \overline{B}$

*therefore,* $x \in A \wedge x \in B$ *(by definition of complement)*

*therefore,* $x \in A \cap B$ *(by definition of intersection)* $\rightarrow A \cap B \neq \emptyset$

*This leads to a contradiction.*

*The* ***second*** *task is to prove that* $A \subseteq \overline{B} \rightarrow A \cap B = \emptyset$

*Again, by way of contradiction, suppose* $A \subseteq \overline{B}$ *and* $A \cap B \neq \emptyset$

*If* $A \cap B \neq \emptyset$ *there exists an* $x$, *such that,* $x \in A \cap B$.

*But,*

$x \in A \cap B$

*therefore,* $x \in A \wedge x \in B$ *(by definition of intersection)*

*therefore,* $x \in A \wedge x \notin \overline{B}$ *(by definition of complement)*$\rightarrow A \nsubseteq \overline{B}$

*This leads to a contradiction.*

*Finally, we have proved:* $A \cap B = \emptyset$ *if and only if* $A \nsubseteq \overline{B}$

- Let A, B, and C be any sets, show that

$$A \times (B \cup C) = (A \times B) \cup (A \times C)$$

$\{(a,b) \mid a \in A,\ b \in (B \cup C)\}$

$A = B$: ① $A \subseteq B$, $B \subseteq A$

② $\forall x \in A \Rightarrow x \in B$

$\forall x \in B \Rightarrow x \in A$

$\forall x \notin A \Rightarrow x \notin B$

*We can prove* $X = Y$ *by showing that if* $a \in X$ *then* $a \in Y$*, and if* $a \notin X$ *then* $a \notin Y$*.*

*Let* $a = (x, y)$*, and suppose* $a \in A \times (B \cup C)$*.*

$(x, y) \in A \times (B \cup C)$

*therefore,* $x \in A \wedge y \in (B \cup C)$ *(by definition of Cartesian Products)*

*therefore,* $x \in A \wedge (y \in B \vee y \in C)$ *(by definition of union)*

*therefore,* $(x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)$ *(by distributive law)*

*therefore,* $(x, y) \in A \times B \vee (x, y) \in A \times C$ *(by definition of Cartesian Products)*

*therefore,* $(x, y) \in (A \times B) \cup (A \times C)$ *(by definition of union)*

*Therefore,* $a \in (A \times B) \cup (A \times C)$

*Let* $a = (x, y)$*, and suppose* $a \notin A \times (B \cup C)$*.*

$= (x, y) \notin A \times (B \cup C)$

*therefore,* $x \notin A \vee y \notin (B \cup C)$ *(by definition of Cartesian Products)*

*therefore,* $x \notin A \vee (y \notin B \wedge y \notin C)$ *(by definition of union)*

*therefore,* $(x \notin A \vee y \notin B) \wedge (x \notin A \vee y \notin C)$ *(by distributive law)*

*therefore,* $(x, y) \notin A \times B \wedge (x, y) \notin A \times C$ *(by definition of Cartesian Products)*

*therefore,* $(x, y) \notin (A \times B) \cup (A \times C)$ *(by definition of union)*

*Therefore,* $a \notin (A \times B) \cup (A \times C)$

*We have finally proved that* $A \times (B \cup C) = (A \times B) \cup (A \times C)$

# Functions and Sequences

# Functions

- Show that the function f(x) = |x| from the set of real numbers to the set of nonnegative real numbers is not invertible, but if the domain is restricted to the set of non-negative real numbers, the resulting function is invertible.

$\mathbb{R} \to \mathbb{R}^+$

$\mathbb{R}^+ \to \mathbb{R}^+$

**Answer**:

For a function to be invertible, it needs to be bijective. Therefore, we need to check if this is one-to-one (injective) and onto (surjective).

(a) Injective: No:

For some $x_1 \neq x_2$, say $x_1 = -x_2$, we have $f(x_1) = f(x_2)$ , by definition of injective, the function is not injective.

If the domain is restricted to the set of nonnegative real numbers, f(x) is injective because

Assume $f(x_1) = f(x_2)$, we have $x_1 = x_2$

Therefore, on the restricted domain f(x) is injective.

(b).Surjective

For some element $b \in rng(f)$, that $b = |a|$, with $a \in R$, b must be positive. Thus, the range is the set of all nonnegative real numbers. Because the range and codomain are the same, we can conclude that f is surjective.

(c) Bijective: No, because it is not injective. Though, on the restricted domain, it is bijective because it is both injective and surjective.

d) Invertible: Again, only on the restricted domain.

- How can we produce the terms of a sequence if the first 10 terms are 1,3,4,7,11,18,29,47,76,123?

- How can we produce the terms of a sequence if the first 10 terms are 1,3,4,7,11,18,29,47,76,123?

- Solution:

$L_n = L_{n-1} + L_{n-2}$, with initial condition $L_1 = 1$, $L_2 = 3$

# Additional exercises on the textbook

- Section 2.1 : 7 11 13 21-27 33 39 43
- Section 2.2 : 5-10 21-23 29-31 37-43
- Section 2.3:  7,15,23-27, 33,35,41,45-47,53,59,71,73
- Section 2.4: 1-4,6-7,9